Your grade: 95.23%

Your latest: **95.23%** • Your highest: **95.23%** • To pass you need at least 80%. We keep your highest score.

Next item

1. According to the lecture, which **best defines** a “community” in a network? 1 / 1 point

- A set of nodes that are all fully disconnected from each other
- A group of nodes arranged in a perfect circle
- A single node that has no edges to any other node
- A group of nodes with many edges among themselves and fewer edges to other groups

Nice work

Correct! This matches the concept of community structure described in the lecture.

2. Based on the examples discussed in this module, e.g. karate club, school friendships, and political blogs, which **real-world factors** often shape community structure? (Select **all** correct answers.) 1 / 1 point

- [ ] The shape or layout used by the network’s visualization software
- [x] Common ideological alignment or affiliation

Nice work

Correct. The political blog example highlighted how ideology drives clustering.

- [x] Shared social or demographic attributes (e.g., age, ethnicity)

Nice work

Correct. The middle/high school friendship network demonstrated this.

- [ ] The random placement of nodes on a graph visualization

3. Which of the following **are** reasons the basic intuitive definition of “many edges within, few edges across” may lead to **trivial or unhelpful** partitions? (Select **all** correct answers.) 1 / 1 point

- [x] One could isolate a single node, minimizing cross-group edges yet yielding no meaningful structure

Nice work

Correct. Putting a single node in one community and all others in another community makes this a trivial split that can happen.

- [ ] Networks are typically too small for such definitions to matter
- [ ] It automatically maximizes the degree of all nodes in each community
- [x] It ignores random chance, not accounting for the expected number of edges across communities

Nice work

Correct. We need to consider the probability of edges appearing under random assignment.

4. Why is a **trivial partition** (like isolating a single node in the karate club) scored **low** by modularity? 1 / 1 point

- Because it makes the network impossible to visualize
- Because I (actual fraction of within-community edges) is similar to **IEXP**, offering no improvement over random
- Because it mistakenly merges both communities into a single group
- Because every node ends up with the same degree

Nice work

Correct! Trivial partitions don’t outperform random expectations, hence low modularity.

5. In the **karate club** example, why did two distinct groups emerge in the network? 1 / 1 point

- A real dispute led members to align either with the instructor (node 1) or the assistant (node 34)
- Nodes were placed far apart on a visual layout
- High-degree nodes disconnected from the network to lower overall density
- Nodes were labeled with different colors at random

Nice work

Correct! The social disagreement drove the network to form two clusters.

6. Why can’t we simply **maximize modularity** by checking **all possible partitions** in a large network? (Select **all** correct answers.) 0.7 / 1 point

- [ ] Because only certain partitions are allowed in community detection
- [x] The number of possible partitions is extremely large, making exhaustive search computationally infeasible

Nice work

Correct. Modularity optimization is NP-hard, requiring heuristics for large networks.

- [ ] Because modularity only applies to networks with a single community

You didn't select all the correct answers

7. Which **factors** does modularity consider when evaluating a partition? (Select **all** correct answers.) 1 / 1 point

- [ ] The absolute position of nodes on a 2D plot
- [ ] The alphabetical order of node labels
- [x] The total number of edges within the identified communities

Nice work

Correct. II measures how many edges fall within each community.

- [x] The expected number of edges within communities if edges were randomly assigned (while keeping degrees fixed)

Nice work

Correct. IEXP captures the random baseline for within-community edges.